

Computer Architectures
Exam of June 25, 2026
Part 1 – possible answers

Exe #2A, Point #3

Assuming that the processor uses 32 bit addresses, each instruction is 4 byte wide, and the BTB is composed of 16 entries, report a table showing the final content of each entry of the BTB and the prediction done for each of the 4 branch instructions if

- The BTB is initially empty (i.e., full of 0s)
- The following instructions are executed in sequence
 1. `add t1, t2, t3` located at the address 0x00002000
 2. `beq t2, t5, l1` located at the address 0x00002004; the branch is taken, and the branch target address is 0x00040024
 3. `addi t1, t0, #1` located at the address 0x00040024
 4. `bne t3, t4, l2` located at the address 0x00040028; the branch is taken, and the branch target address is 0x000AA004
 5. `sll t1, t0, #2` located at the address 0x000AA004
 6. `sub t2, t0, t5` located at the address 0x000AA008
 7. `beq t2, t3, l3` located at the address 0x000AA008; the branch is not taken
 8. `beq t2, t5, l4` located at the address 0x000AA010; the branch is taken, and the branch target address is 0x00002000
 9. `add t1, t2, t3` located at the address 0x00002000
 10. `beq t2, t5, l1` located at the address 0x00002004; the branch is taken, and the branch target address is 0x00040024.

Exe 2: BTB initial content

[illegible]

Exe #2: BTB final content

address	target
0x00000000	0x00000000
0x00002004	0x00040024
0x00000000	0x00000000
0x00000000	0x00000000
0x000AA010	0x00002000
0x00000000	0x00000000
0x00000000	0x00000000
0x00000000	0x00000000
0x00000000	0x00000000
0x00000000	0x00000000
0x00000000	0x00000000
0x00040028	0x000AA004
0x00000000	0x00000000
0x00000000	0x00000000
0x00000000	0x00000000
0x00000000	0x00000000

Exe #2B, Point #3

- Assuming that the processor uses 32 bit addresses, each instruction is 4 byte wide, and the BTB is composed of 16 entries, report a table showing the final content of each entry of the BTB and the prediction done for each of the 4 branch instructions if
 - The BTB is initially empty (i.e., full of 0s)
 - The following instructions are executed in sequence
 1. `sub t3, t5, t4` located at the address 0x000A4004
 2. `beq t2, t5, l1` located at the address 0x000A4008; the branch is taken, and the branch target address is 0x00080024
 3. `addi t2, t0, #1` located at the address 0x00080024
 4. `bne t3, t4, l2` located at the address 0x00080028; the branch is taken, and the branch target address is 0x000BB004
 5. `sll t1, t0, #2` located at the address 0x000BB004
 6. `sub t1, t0, t3` located at the address 0x000BB008
 7. `beq t2, t3, l3` located at the address 0x000BB00C; the branch is not taken
 8. `beq t2, t5, l4` located at the address 0x000BB010; the branch is taken, and the branch target address is 0x000A4004
 9. `add t1, t2, t3` located at the address 0x000A4004
 10. `beq t2, t5, l1` located at the address 0x000A4008; the branch is taken, and the branch target address is 0x00080024.

Exe #2B: BTB initial content

[illegible]

Exe #2B: BTB final content

address	target
0x00000000	0x00000000
0x00000000	0x00000000
0x000A4008	0x00080024
0x00000000	0x00000000
0x000BB010	0x000A4004
0x00000000	0x00000000
0x00000000	0x00000000
0x00000000	0x00000000
0x00000000	0x00000000
0x00000000	0x00000000
0x00000000	0x00000000
0x00080028	0x000BB004
0x00000000	0x00000000
0x00000000	0x00000000
0x00000000	0x00000000
0x00000000	0x00000000
0x00000000	0x00000000